TIMOTHY JOHNSTONE

Sr. Principal Computational Scientist | AWS-Certified Cloud Solutions Architect

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PROFESSIONAL SUMMARY

Ph.D. geneticist and computational leader specializing in cloud architecture and NGS bioinformatics. 20+ years of experience analyzing data from novel sequencing assays, developing high-throughput screening modalities, and engineering cloud-native bioinformatics pipelines. Passionate about building and managing secure, scalable cloud systems and developer resources from the ground up to accelerate high-growth biotech organizations and analytical teams.

AREAS OF EXPERTISE

Infrastructure & Cloud

- Multi-account AWS (Control Tower)
- Scientific Computing (HPC)
- Infrastructure as Code (IaC)
- Cloud Governance & FinOps

Computational Biology

- NGS/HTS Data Analysis
- Data Management
- Pipeline Engineering (Nextflow)
- RNA, Cell, & Gene Therapy

Development & AI

- Python, R, Unix/Bash
- CI/CD (GitHub Actions, Docker)
- Generative AI, LLM, RAG, MLOps
- Version Control & Documentation

PROFESSIONAL EXPERIENCE

Sr. Principal Computational Scientist; Cloud Architect | **Shape Therapeutics** | Seattle, WA | 2021 – 2025

Head of Cloud and technical lead for a division of computational biology and ML scientists, managing all aspects of cloud infrastructure. Oversaw computational analysis of complex genomic screens and guided the development, documentation practices, and culture of a successful team.

- **AWS Infrastructure:** Built & managed a multi-account AWS environment for 120+ users and 200TB+ of data from scratch, including cost optimization, networking, security, systems administration, authentication, and user training
- **AI/MLOps**: Orchestrated the development of an internal LLM+RAG chatbot for researchers and deployed GPU compute, ML experiment tracking, and infrastructure for protein engineering and foundation model training
- **Automation & Data Management:** Engineered a fully automated, Python-based solution for synchronizing Illumina and Nanopore NGS data to the cloud, and orchestrated internal and external data sharing for collaborations
- **CI/CD & DevOps:** Built robust CI/CD pipelines using GitHub Actions and a secure portal (AWS ALB, ECS, S3, VPC, IAM, Route53, CloudFormation, Okta) for internally hosted web-apps, dashboards, static hosting, and Docker images
- **Software Development:** Developed a Python package for optimized fuzzy matching of DNA/RNA sequences which was integrated into many internal analytical pipelines, reducing development and primary analysis time over 50%
- **Bioinformatics:** Designed experiments, authored Nextflow pipelines, and processed NGS data for novel high-throughput screening assays to improve editing efficiency, assess viral genome integrity, and refine cell lines

Principal Scientist | **Bristol Myers Squibb (Celgene)** | Seattle, WA | 2019 – 2021

Drove the development of novel analytical assays for CAR/TCR constructs, executed multi-omics analyses, developed analytical applications and pipelines, and served as the primary lead for a major cloud migration.

- **Assay Development:** Led the analysis of lentiviral integration site analysis data and CRISPR indels, resulting in non-clinical reports that directly informed multiple successful BLA and IND filings
- **Pipeline Engineering:** Developed rapid analytical pipelines for Nanopore sequencing data to determine recombination patterns in CAR and eTCR constructs, leading to integrity improvement for multiple clinical candidates
- **Cloud Strategy:** Planned and executed a major post-acquisition cloud migration to ensure the scalability and integration of 3+ years of NGS analytics data, AI/ML efforts, web applications, and reproducible research tools
- **Web Applications:** Developed an R Shiny application used to prioritize candidates for preclinical CAR development

Senior Data Scientist | **Juno Therapeutics** | Seattle, WA | 2016 – 2019

Leveraged breakthrough single-cell and bulk sequencing technologies to discover novel antigen receptors for cancer immunotherapy. Constructed bioinformatics tools and developed the company's cloud-based computing infrastructure for single-cell and bulk sequencing technologies. Juno was acquired by Celgene during my tenure.

- **NGS Pipelines:** Constructed and optimized pipelines for ATACSeq, RNASeq, and TMISeq to yield actionable biological conclusions, collaborating with molecular biology and translational researchers throughout the company
- **Clonality Analysis:** Led the development of timecourse T-cell clonality analyses in R, tracking patient samples from product to post-infusion. Refined algorithms to improve signal-to-noise ratios and eliminate false positives
- **Process Optimization:** Implemented event notifications and automations to improve the efficiency of the NGS sequencing workflow for data scientists, eliminating hours to days of lost communication time with each analysis
- **Cloud Development & HPC:** Managed the company's NGS analytics cloud infrastructure and optimized Snakemake pipelines on autoscaling clusters (AWS ParallelCluster) with thousands of CPUs

Graduate Researcher | **Yale University School of Medicine** | New Haven, CT | 2012-2016

Investigated non-canonical translation and regulatory roles of uORFs in the vertebrate transcriptome. Utilized ribosome profiling, GRO-seq, microRNA-seq, and others to discover translation in RNAs previously deemed non-coding, find novel enhancers in viral response, and elucidate piRNA biogenesis. Developed custom algorithms/visualizations in Python and R.

Undergrad Team Leader | **Brown Center for Computational Molecular Biology** | Providence RI | 2009-2012

Managed a team of undergraduate lab members as lead annotator for the discovery of DNA cis-regulatory modules and release of a genome browser focused on transcriptional regulation.

Bioinformatics Research Assistant | **OHSU Affymetrix Microarray Core** | Portland, OR | 2006; 2008

Performed statistical analyses, managed large scale blood sample study data, developed protocols for nanogram-level RNA handling, and acted as a biological advisor for PSU physics labs, resulting in authorship on multiple publications.

EDUCATION

Ph.D. in Genetics, Yale University, *New Haven, CT* | 2012-2016

B.S. with Honors in Computational Biology, Brown University, *Providence, RI* | 2008-2012

For publications, patents, certifications, and additional qualifications, visit ***linkedin.com/in/timjohnstone***